

Limitations

SQLite does not implement the complete SQL 92 standard and hence it has some limitations:

- ALTER: Altering the table is not completely supported. Only renaming and adding columns to the table is possible, which makes sense as the schema remains the same for the lifetime of the embedded application. As the database is not expected to be large, to overcome this limitation, a new database can be created and data can be copied.
- JOIN: Only the left outer join is supported. The unsupported right outer join can be done by executing a left outer join on the reversed list of tables in pairs.
- Triggers: Triggers are supported only for rows.
- Views: The *Insert*, *Update* and *Delete* operations are not permitted on views in SQLite, which can be overcome by triggers. Search the SQLite website for more details on this limitation.
- SQLite is optimised for performance by the imposition of certain limits, both at compile time and runtime, like BLOB size, number of columns, number of tables in a join, etc.

Older versions of SQLite may have more limitations, like foreign key constraints are supported only from 3.6.19, etc.

Alternatives

Some of the alternatives, though not as lightweight as SQLite, are:

- HSQldb: Written in Java, and so requires JRE.
- Firebird: A cross-platform RDB that has full SQL92 conformance; it is lightweight, but not an embedded database.
- Ultralite: A cross-platform commercial product from Sybase. 

References

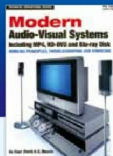
- <http://www.sqlite.org>
- <http://www.if-not-true-then-false.com/2012/php-pdo-sqlite3-example/>
- A complete comparison of the known databases are maintained at http://en.wikipedia.org/wiki/Comparison_of_relational_database_management_systems
- Android tutorial on SQLite <http://www.linuxforu.com/2010/12/android-application-development-playing-with-sqlite-database/>

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